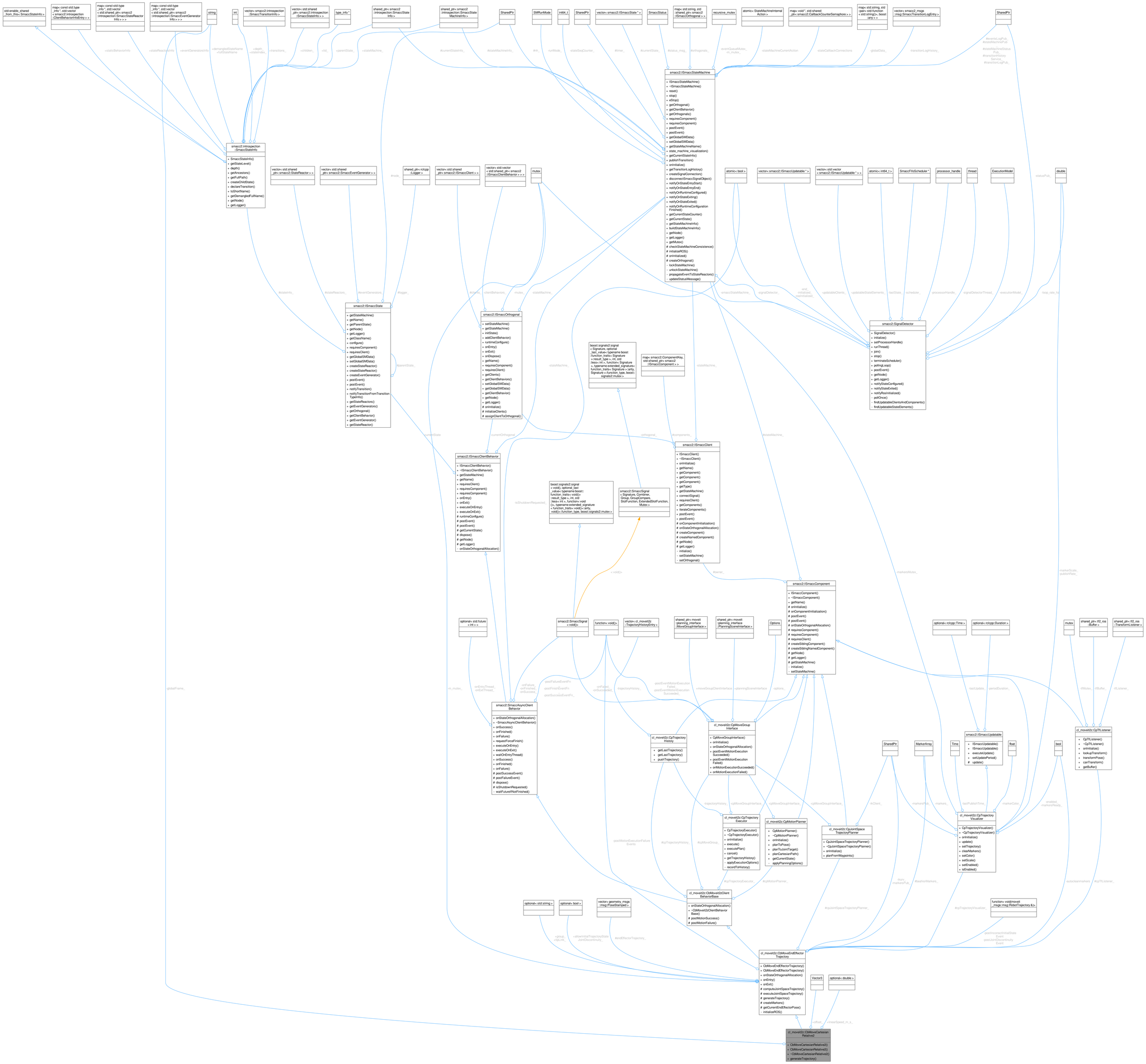
std::enable_shared
_from_this< SmaccStateInfo >
map< const std::type
_info *, std::vector
< smacc2::introspection
::ClientBehaviorInfoEntry > >
map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccStateReactor
Info > > >
map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccEventGenerator
Info > > >
string
int
vector< smacc2::introspection
::SmaccTransitionInfo >
vector< std::shared
_ptr< smacc2::introspection
::SmaccStateInfo > >
type_info *
shared_ptr< smacc2
::introspection::SmaccState
Info >
shared_ptr< smacc2
::introspection::SmaccState
MachineInfo >
SharedPtr
SMRunMode
int64_t
SharedPtr
vector< smacc2::ISmaccState * >
SmaccStatus
map< std::string, std
::shared_ptr< smacc2
::ISmaccOrthogonal > >
recursive_mutex
atomic< StateMachineInternal
Action >
map< void *, std::shared
_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std
::pair< std::function
< std::string()>, boost
::any > >
vector< smacc2_msgs
::msg::SmaccTransitionLogEntry >
SharedPtr
+staticBehaviorInfo
+stateReactorsInfo_
+eventGeneratorsInfo
+demangledStateName
+fullStateName
+depth_
+stateIndex_
+transitions_
+children_
+id_
+parentState_
+stateMachine_
#currentStateInfo_
#stateMachineInfo_
#in_
-runMode_
-stateSeqCounter_
#timer_
#currentState_
#status_msg_
#orthogonals_
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
#eventsLogPub_
#stateMachinePub
#stateMachineStatus
Pub_
#transitionHistory
Service_
#transitionLogPub_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration
Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
smacc2::introspection
::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
vector< std::shared
_ptr< smacc2::StateReactor > >
vector< std::shared
_ptr< smacc2::SmaccEventGenerator > >
#node_
shared_ptr< rclcpp
::Logger >
vector< std::shared
_ptr< smacc2::ISmaccClient > >
vector< std::vector
< std::shared_ptr< smacc2
::ISmaccClientBehavior > > >
mutex
atomic< bool >
vector< smacc2::ISmaccUpdatable * >
vector< std::vector
< smacc2::ISmaccUpdatable * > >
atomic< int64_1 >
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
-statusPub_
double
#stateInfo_
#stateReactors_
#eventGenerators_
#logger_
#clients_
#clientBehaviors_
-mutex_
-stateMachine_
-smaccStateMachine_
-signalDetector_
-end_
-initialized_
-wasInitialized_
-updatableClients_
-updatableStateElements_
-lastState_
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
-loop_rate_hz
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition
TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
boost::signals2::signal
< Signature, optional
_last_value< typename boost
::function_traits< Signature
>::result_type >, int, std
::less< int >, function< Signature
>, typename extended_signature<
function_traits< Signature >::arity,
Signature >::function_type, boost::
signals2::mutex >
map< smacc2::ComponentKey,
std::shared_ptr< smacc2
::ISmaccComponent > >
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
-currentState
-currentOrthogonal
-orthogonal_
#components_
#stateMachine_
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
-isShutdownRequested_
boost::signals2::signal
< void(), optional_last
_value< typename boost::
function_traits< void()>
::result_type >, int, std
::less< int >, function< void
()>, typename extended_signature
< function_traits< void()>::arity,
void()>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal
< Signature, Combiner,
Group, GroupCompare,
SlotFunction, ExtendedSlotFunction,
Mutex >
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
< void()>
#owner_
-markersMutex_
-markerScale_
-publishRate_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
optional< std::future
< int > >
smacc2::SmaccSignal
< void()>
function< void()>
vector< cl_moveit2z
::TrajectoryHistoryEntry >
shared_ptr< moveit
::planning_interface
::MoveGroupInterface >
shared_ptr< moveit
::planning_interface
::PlanningSceneInterface >
Options
optional< rclcpp::Time >
optional< rclcpp::Duration >
mutex
shared_ptr< tf2_ros
::Buffer >
shared_ptr< tf2_ros
::TransformListener >
-globalFrame_
-in_mutex_
-onEntryThread_
-onExtThread_
-onFailure_
-onFinished_
-onSuccess_
-postFailureEventFn
-postFinishEventFn
-postSuccessEventFn_
-onFailed_
-onSucceeded_
-trajectoryHistory_
-postEventMotionExecution
Failed_
-postEventMotionExecution
Succeeded_
+moveGroupClientInterface
+planningSceneInterface
-options_
-lastUpdate_
-periodDuration_
-tfMutex_
-tfBuffer_
-tfListener_
smacc2::SmaccAsyncClient
Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
cl_moveit2z::CpTrajectory
History
+ getLastTrajectory()
+ getLastTrajectory()
+ pushTrajectory()
cl_moveit2z::CpMoveGroup
Interface
+ CpMoveGroupInterface()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ postEventMotionExecution
Succeeded()
+ postEventMotionExecution
Failed()
+ onMotionExecutionSucceeded()
+ onMotionExecutionFailed()
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
cl_moveit2z::CpTIListener
+ CpTIListener()
+ ~CpTIListener()
+ onInitialize()
+ lookupTransform()
+ transformPose()
+ canTransform()
+ getBuffer()
SharedPtr
MarkerArray
Time
float
bool
-trajectoryHistory_
-cpMoveGroupInterface_
-cpMoveGroupInterface_
-cpMoveGroupInterface_
-ikClient_
-markersPub_
-markers_
-lastPublishTime_
-markerColor_
-enabled_
-markersReady_
cl_moveit2z::CpTrajectory
Executor
+ CpTrajectoryExecutor()
+ ~CpTrajectoryExecutor()
+ onInitialize()
+ execute()
+ executePlan()
+ cancel()
+ getTrajectoryHistory()
- applyExecutionOptions()
- recordToHistory()
cl_moveit2z::CpMotionPlanner
+ CpMotionPlanner()
+ ~CpMotionPlanner()
+ onInitialize()
+ planToPose()
+ planToJointTarget()
+ planCartesianPath()
+ getCurrentState()
- applyPlanningOptions()
cl_moveit2z::CpJointSpace
TrajectoryPlanner
+ CpJointSpaceTrajectoryPlanner()
+ ~CpJointSpaceTrajectoryPlanner()
+ onInitialize()
+ planFromWaypoints()
cl_moveit2z::CpTrajectory
Visualizer
+ CpTrajectoryVisualizer()
+ ~CpTrajectoryVisualizer()
+ onInitialize()
+ update()
+ setTrajectory()
+ clearMarkers()
+ setColor()
+ setScale()
+ setEnabled()
+ isEnabled()
-postMotionExecutionFailure
Events
#cpTrajectoryHistory_
#cpMoveGroup_
-cpTrajectoryExecutor_
-cpMotionPlanner_
#cpJointSpaceTrajectoryPlanner_
-ikSolv
-markersPub_
#beahiorMarkers_
#cpTrajectoryVisualizer_
-autocleanmarkers
#cpTIListener_
cl_moveit2z::CbMoveIt2rClient
BehaviorBase
+ onStateOrthogonalAllocation()
+ ~CbMoveIt2rClientBehavior
Base()
# postMotionSuccess()
# postMotionFailure()
optional< std::string >
optional< bool >
vector< geometry_msgs
::msg::PoseStamped >
+group_
+ikLink_
+allowInitialTrajectoryState
JointDiscontinuity_
#endEffectorTrajectory_
cl_moveit2z::CbMoveEndEffector
Trajectory
+ CbMoveEndEffectorTrajectory()
+ CbMoveEndEffectorTrajectory()
+ onStateOrthogonalAllocation()
+ onEntry()
+ onExit()
# computeJointSpaceTrajectory()
# executeJointSpaceTrajectory()
# generateTrajectory()
# createMarkers()
# getCurrentEndEffectorPose()
- initializeROS()
function< void(moveit
_msgs::msg::RobotTrajectory &)>
-postIncorrectInitialState
Event_
-postJointDiscontinuity
Event_
Vector3
optional< double >
+offset_
+linearSpeed_m_s_
cl_moveit2z::CbMoveCartesian
Relative2
+ CbMoveCartesianRelative2()
+ CbMoveCartesianRelative2()
+ ~CbMoveCartesianRelative2()
+ generateTrajectory()